

1. Healthcare: Spatial Distribution of Respiratory Issues vs. Air Quality

- **Background:** Urbanization has led to localized pockets of high air pollution.
- **Problem Statement:** Is there a statistically significant correlation between specific neighborhood air quality index (AQI) readings and respiratory emergency admissions?
- **Research Gap:** Most studies look at city-wide averages; this project targets granular, street-level correlations.
- **Expected Dataset:** Open Government Data (City AQI) + Public Health Department records.
- **Future Scope:** Integrate with Deep Learning for time-series forecasting of hospital admissions.

2. Agriculture: Soil Moisture Depletion vs. Crop Yield Efficiency

- **Background:** Water scarcity is the primary threat to sustainable farming.
- **Problem Statement:** How does the rate of soil moisture depletion differ across various crop types under identical irrigation conditions?
- **Research Gap:** Lack of comparative analysis on water retention efficiency between cash crops and staple grains.
- **Expected Dataset:** UCI Machine Learning Repository (Agricultural Datasets).
- **Future Scope:** Develop an IoT-based predictive irrigation model.

3. Smart Cities: The "Last Mile" Bottleneck in Public Transport

- **Background:** Public transit systems are efficient, but the "last mile" connectivity remains a barrier.
- **Problem Statement:** Identify high-congestion nodes where commuters abandon public transport for private vehicles.
- **Research Gap:** Most models focus on traffic volume; this focuses on commuter *behavioral switches*.
- **Expected Dataset:** City Transit Authority API data / Google Maps mobility data.
- **Future Scope:** Optimization algorithms for bus route adjustments.

4. Climate Change: Urban Heat Island (UHI) Effect Intensity

- **Background:** Cities are significantly warmer than surrounding rural areas.
- **Problem Statement:** Quantify the temperature variance between dense concrete zones and green spaces in your city using historical weather data.
- **Research Gap:** UHI is often studied globally; localized, specific-city micro-climate quantification is rare.
- **Expected Dataset:** NASA EarthData / NOAA historical weather records.
- **Future Scope:** Develop an Explainable AI (XAI) model to predict cooling needs.

5. Education Analytics: Digital Engagement vs. Assessment Outcomes

- **Background:** The transition to hybrid learning has altered student study patterns.
- **Problem Statement:** Is there a correlation between the time-of-day of student engagement on learning management systems and academic performance?

- **Research Gap:** Most studies are qualitative; this requires quantitative correlation analysis between timestamped logins and scores.
- **Expected Dataset:** Kaggle (Student Performance / Learning Datasets).
- **Future Scope:** Early warning systems for at-risk students using Reinforcement Learning.

6. Cyber Security: Network Traffic Packet Length Distribution

- **Background:** Malicious traffic often exhibits different packet size distributions than normal traffic.
- **Problem Statement:** Can we identify statistical signatures of non-standard traffic without using ML, purely by analyzing packet length frequency distributions?
- **Research Gap:** Using basic descriptive statistics to identify "pre-anomalies" before applying heavy AI.
- **Expected Dataset:** NSL-KDD or University of New Brunswick (UNB) Intrusion Detection data.
- **Future Scope:** Anomaly detection systems using Federated Learning.

7. Finance: Inflationary Impact on Retail Consumer Basket Composition

- **Background:** Prices fluctuate, but consumer priorities change non-linearly.
- **Problem Statement:** How does the consumer "luxury vs. necessity" ratio change when inflation exceeds a specific threshold?
- **Research Gap:** Linking macroeconomic data (CPI) to micro-consumer spending behavior.
- **Expected Dataset:** World Bank Open Data / National Statistical Bureau.
- **Future Scope:** Financial forecasting models using Time-Series Analysis.

8. Banking: Loan Approval Bias Analysis

- **Background:** Financial inclusion is a major SDG goal.
- **Problem Statement:** Conduct a statistical parity audit on bank loan approval data to identify demographic biases in lending.
- **Research Gap:** Identifying *hidden* variables that correlate with approval, even if protected classes are masked.
- **Expected Dataset:** UCI Statlog (German Credit Data).
- **Future Scope:** Algorithmic Fairness auditing frameworks.

9. Retail: Customer Churn Propensity via Interaction Density

- **Background:** Customer retention is cheaper than acquisition.
- **Problem Statement:** Does the frequency of customer service interactions serve as a leading indicator of churn before the customer actually leaves?
- **Research Gap:** Moving from static churn prediction to dynamic interaction-based churn triggers.
- **Expected Dataset:** Kaggle (Telco Customer Churn).
- **Future Scope:** Predictive maintenance for customer relationships.

10. Supply Chain: Logistics Delay Causality in E-commerce

- **Background:** Delivery delays are the #1 dissatisfaction point in e-commerce.
- **Problem Statement:** Is there a seasonal "delay profile" for specific logistics hubs, and does it correlate with regional weather patterns?
- **Research Gap:** Multimodal causal analysis (Weather + Geography + Operations).
- **Expected Dataset:** Olist E-commerce Dataset.
- **Future Scope:** Digital Twins for supply chain optimization.

11. Energy: Household Load Profile Clustering

- **Background:** Smart grids need to balance demand.
- **Problem Statement:** Can we group households into "usage clusters" based on hourly consumption spikes?
- **Research Gap:** Unsupervised discovery of household behavioral archetypes.
- **Expected Dataset:** UCI Household Electric Power Consumption.
- **Future Scope:** Energy-saving policy recommendations.

12. Electric Vehicles (EVs): Charging Infrastructure Optimization

- **Background:** EV adoption is limited by "range anxiety."
- **Problem Statement:** Analyze the optimal placement of charging stations based on traffic density and residential parking data.
- **Research Gap:** Using spatial analysis to minimize "stranded time."
- **Expected Dataset:** OpenStreetMap Data.
- **Future Scope:** Multi-agent systems for station management.

13. Smart Manufacturing: Sensor Drift in Production Lines

- **Background:** Machines degrade over time.
- **Problem Statement:** Perform trend analysis on sensor logs to detect if variance in output quality correlates with equipment age.
- **Research Gap:** Predictive maintenance without needing massive labeled failure data.
- **Expected Dataset:** NASA Turbofan Engine Degradation Dataset.
- **Future Scope:** Anomaly detection using Autoencoders.

14. Social Media Analytics: Sentiment Polarity of Public Health Crises

- **Background:** Misinformation spreads faster than truth.
- **Problem Statement:** Analyze the evolution of public sentiment regarding a vaccine or health mandate over time.
- **Research Gap:** Quantifying the "panic index" in social discourse.
- **Expected Dataset:** Twitter/Reddit Archives (Kaggle).
- **Future Scope:** Knowledge Graphs for rumor propagation.

15. Transportation: The Impact of Traffic Signals on Carbon Footprint

- **Background:** Idling cars produce the most emissions.
- **Problem Statement:** Calculate the estimated CO2 emission per intersection based on average wait times.
- **Research Gap:** Intersection-level carbon footprint quantification.

- **Expected Dataset:** DOT Traffic Data.
- **Future Scope:** Reinforcement Learning for traffic light control.

16. Urban Planning: Walkability and Local Economic Vitality

- **Background:** Walkable cities thrive.
- **Problem Statement:** Is there a correlation between street walkability scores and the number of small businesses per square kilometer?
- **Research Gap:** Mapping the "human-scale" economic health of a city.
- **Expected Dataset:** OpenStreetMap + Census Bureau.
- **Future Scope:** Urban development planning tools.

17. Satellite Data: Deforestation Rate vs. Road Expansion

- **Background:** Road networks are the primary vector for deforestation.
- **Problem Statement:** Quantify the distance of deforestation events from the nearest major road network.
- **Research Gap:** Spatial-temporal causality in environmental degradation.
- **Expected Dataset:** Global Forest Watch / Satellite Imagery API.
- **Future Scope:** Computer Vision for satellite image classification.

18. Tourism: Seasonal Demand vs. Climate Variability

- **Background:** Tourism is highly sensitive to weather.
- **Problem Statement:** Determine if tourist inflow to coastal areas is resilient to anomalous weather events.
- **Research Gap:** Climate change resilience indexing for tourism-dependent economies.
- **Expected Dataset:** UNWTO Data / Local Tourism Board.
- **Future Scope:** Recommendation systems with environmental constraints.

19. Government: Public Service Delivery Lag Analysis

- **Background:** Bureaucratic friction is a major global issue.
- **Problem Statement:** Analyze the time taken for file movement between government departments and identify the "bottleneck" department.
- **Research Gap:** Quantifying bureaucratic inefficiency via process mining.
- **Expected Dataset:** Open Government Data (RTI or public records).
- **Future Scope:** Optimization of public service workflows.

20. Digital Health: Wearable Data Reliability Analysis

- **Background:** Consumer health data is often noisy.
- **Problem Statement:** Compare the data consistency between different wearable devices for the same user activity (e.g., heart rate monitoring).
- **Research Gap:** Establishing a "noise floor" for consumer-grade medical devices.
- **Expected Dataset:** Kaggle (MHealth Dataset).
- **Future Scope:** Signal denoising using Deep Learning.

21. Environmental Sustainability: Plastic Waste Composition Analysis

- **Background:** Waste sorting is inefficient.
- **Problem Statement:** Analyze regional waste audit data to identify the most frequent non-recyclable items.
- **Research Gap:** Data-driven policy suggestions for local waste management.
- **Expected Dataset:** EPA Waste Management Data.
- **Future Scope:** Computer Vision for automated waste sorting.

22. Public Safety: Temporal-Spatial Crime Clustering

- **Background:** Crime is not uniformly distributed.
- **Problem Statement:** Identify "hot time" and "hot spot" crime patterns. Is there a correlation with public lighting availability?
- **Research Gap:** Causal link between urban infrastructure and crime prevention.
- **Expected Dataset:** FBI Uniform Crime Reporting (UCR) / Local Police Open Data.
- **Future Scope:** Predictive Policing Ethics and Fairness.

23. Precision Medicine: Genomic Data Correlation with Drug Response

- **Background:** Not all drugs work for all patients.
- **Problem Statement:** Identify statistical trends in patient health outcomes based on genotype/drug-dosage pairs.
- **Research Gap:** Finding outliers in treatment response.
- **Expected Dataset:** TCGA (The Cancer Genome Atlas).
- **Future Scope:** Personalized medicine systems using Neural Networks.

24. Energy: Grid Peak-Load Prediction

- **Background:** Renewable energy is intermittent.
- **Problem Statement:** Analyze historical load data to identify the "reliability window" of renewable sources versus demand spikes.
- **Research Gap:** Grid stability analysis for high-renewables scenarios.
- **Expected Dataset:** Entsoe Transparency Platform.
- **Future Scope:** Storage optimization using Reinforcement Learning.

Submission Template

Students must provide a **single GitHub repository** containing:

1. **/data:** Raw and cleaned CSV/JSON files.
2. **/src:** Python scripts (`main.py`, `analysis.py`, `viz.py`).
3. **/notebooks:** Exploratory Jupyter Notebooks (`eda_analysis.ipynb`).
4. **/docs:**
 - `Final_Report.pdf` (IEEE Format).
 - `Presentation.pptx`.
 - `Video_Demo.mp4` (5-minute walkthrough)
5. `README.md`: Instructions on how to run the project.